higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL WORKMANSHIP
NQF LEVEL 3

(12041013)

19 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Read ALL the questions carefully.
2. Answer ALL the questions.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.4) in the ANSWER BOOK.

1.1.1 The process of setting goals to achieve specific deadlines:

A A task list
B A bar chart
C Time management
D A safety factor

1.1.2 The workplace condition caused by sexual harassment or production pressure:

A Extreme heat
B Noise
C Poor lighting
D Stress

1.1.3 The safety procedure used to inform workers that an electrical circuit has been isolated due to work in progress:

A Maintenance
B Lockout and tagout
C Isolation verification
D Permit work

1.1.4 Which process is used to ensure the safety of workers in the workplace?

A Supervision
B Planning and scheduling
C Safety policies and procedures
D Training

(4 × 1) (4)

1.2 List THREE factors to consider when planning and scheduling work. (3)

1.3 State TWO reasons why planning and scheduling are so important. (2)

1.4 State TWO objectives of using bar charts. (2)

1.5 Name THREE steps that must be taken to prevent accidents. (3)

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1.6 Employers are responsible for the health and safety of their employees.

1.6.1 Explain THREE ways in which employers can train employees on health and safety awareness in the workplace. (3)

1.6.2 List the THREE steps that employers must follow to prevent injuries and illness in the workplace. (3)

1.7 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.7.1–1.7.5) in the ANSWER BOOK.

1.7.1 Permission to work is not necessary when doing maintenance on electrical equipment.

1.7.2 Late delivery of materials and equipment will not affect work deadlines.

1.7.3 Unexpected incidents can prevent you from keeping to your daily, weekly or monthly plan.

1.7.4 A physical or mental disability is considered as a personal factor.

1.7.5 Careful planning and scheduling will not minimise waste and give workers direction.

(5 × 1) (5)
[25]

QUESTION 2

2.1 There are TWO basic types of electrical test instruments used, namely analogue meters and digital meters.

Name TWO differences between these types of test instruments. (2)

2.2 Name the test instruments used to measure, view or test the following electrical quantities or devices:

2.2.1 The rotational speed of an electric motor rotor or armature

2.2.2 Vibrating or rotating objects so as to make them appear to be stationary

2.2.3 The condition of the insulation between cable conductors or electric motor windings

(3 × 1) (3)

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2.3 Name TWO hazards that can affect the accurate operation of testing and measuring instruments. (2)

2.4 Draw a fully labelled circuit diagram showing how you would connect an ammeter in an electrical circuit using the following components: battery/supply, switch and the load. (5)

2.5 Calculate the voltmeter full-scale deflection if the line voltage of an instrument transformer is 400 V and the primary and secondary windings are 600 and 60, respectively. (3)

[15]

QUESTION 3

3.1 Explain the operation of a mechanical timer used in washing machines. (3)

3.2 List the THREE different types of heaters. (3)

3.3 Complete the following paragraph by writing down the missing words next to the question number (3.3.1–3.3.4) in the ANSWER BOOK.

> The oven thermostat consists of a (3.3.1) … bulb that contains a volatile liquid. The bulb is connected by means of a capillary (3.3.2) … to the expansion bellows inside the switch. The adjustment of the control knob changes the distance between the contacts so that the time delay for make and (3.3.3) … may be decreased or increased depending on the (3.3.4) … (4)

[Source: *Electrical Workmanship NCV Level 3: Jowaheer.*]

3.4 Briefly explain the basic function of each of the following components:

3.4.1 A microwave magnetron

3.4.2 A stove warmer drawer

3.4.3 A tumble dryer

(3 × 1) (3)

3.5 Explain TWO electrical regulation conditions concerning a cooking appliance. (2)

[**15]**

QUESTION 4

4.1 State FIVE precautions when doing testing on electrical appliances and power tools. (5)

4.2 Maintenance should be done on appliances and power tools from time to time to keep them in good working order.

Explain in FOUR points, how you would check or repair the brushes of a portable drilling machine. (4)

4.3 With regard to faulty power tools:

Choose a possible cause from COLUMN B that matches a fault condition in COLUMN A. Write only the letter (A–E) next to the question number (4.3.1–4.3.4) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.3.1	Does not work at all	A	armature or commutator faulty
4.3.2	Motor is sluggish and runs slower than normal	B	motor bearing worn
4.3.3	Machine is very noisy	C	faulty plug
4.3.4	Machine is overheating	D	timer faulty
		E	air vent blocked

(4 × 1) (4)

4.4 With regard to faulty geysers:

Choose a possible cause from COLUMN B that matches a fault condition in COLUMN A. Write only the letter (A–C) next to the question number (4.4.1–4.4.2) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.4.1	No hot water	A	thermostat setting too low
4.4.2	Not enough hot water	B	no seal
		C	faulty element

(2 × 1) (2)

[15]

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QUESTION 5

5.1 Name the type of rotor used in a capacitor-start capacitor-run motor. (1)

5.2 Draw a fully labelled circuit diagram of a capacitor-start induction-run motor. (6)

5.3 A relay is made up of NO contacts as well as NC contacts and other parts or components.

Name TWO other components of a relay. (2)

5.4 Motors are classified according to the type of motor enclosures used.

Explain the term *waterproof protection*. (3)

5.5 5.5.1 What does DOL stand for? (1)

5.1.2 What are the TWO colours used in DOL wiring push buttons? (2)

[15]

QUESTION 6

6.1 Describe the *SMAW process*. (2)

6.2 Explain what you understand by the following terms:

6.2.1 Power supply

6.2.2 Welding cable

6.2.3 Electrode holders

(3 × 1) (3)

6.3 Welding work is a very hazardous process.

Name FOUR major hazards to be aware of when doing welding. (4)

6.4 List FOUR different types of welded joints. (4)

6.5 State TWO types of PPE used during SMAW. (2)

[15]

TOTAL: 100

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